

AURANGABAD, India

WMO: 430140

Lat: 19.863N

Lon: 75.398E

Elev: 583

StdP: 94.52

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	11.1	12.5	2.0	4.7	24.9	3.8	5.3	24.0	5.5	25.5	4.5	25.6	0.1	340	0.314

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	12.5	40.3	22.8	39.4	22.7	38.4	22.5	26.9	35.7	25.9	33.5	25.3	31.9	2.9	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.8	21.3	30.5	24.2	20.5	28.9	23.7	19.9	28.0	88.9	35.7	84.3	33.4	81.4	32.1	31.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.5	5.6	5.0	DB	8.7	42.2	1.4	0.7	7.7	42.7	6.9	43.1	6.2	43.6	5.2	44.1	(4)
(5)			WB	6.2	28.4	1.4	1.5	5.2	29.5	4.4	30.3	3.6	31.1	2.6	32.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	26.2	21.3	24.1	28.0	31.5	32.6	29.0	26.2	25.5	26.0	25.6	23.3	21.3
	DBStd	3.93	2.03	2.23	2.41	2.08	1.82	2.45	1.49	1.27	1.17	1.55	1.80	2.00
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	4	3	0	0	0	0	0	0	0	0	0	0	2
	CDD10.0	5912	351	395	559	646	699	570	501	481	479	484	398	349
	CDD18.3	2876	95	162	301	396	441	320	243	223	229	226	149	93
	CDH23.3	33077	997	1774	3898	6061	6912	3787	1863	1492	1768	2126	1423	977
	CDH26.7	16772	294	831	2226	3883	4533	1890	535	361	545	897	509	267
Wind	WSAvg	1.8	0.9	1.1	1.3	1.7	2.9	3.2	3.0	2.9	1.8	1.1	0.9	0.8
Precipitation	PrecAvg	910	3	3	10	7	26	166	200	210	175	84	18	10
	PrecMax	1399	20	18	97	36	220	497	458	398	340	383	110	101
	PrecMin	591	0	0	0	0	0	43	41	67	65	7	0	0
	PrecStd	198	5	5	21	10	48	93	103	84	73	80	33	25
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.0	35.2	39.0	41.2	41.8	40.0	33.6	32.1	32.9	34.2	32.3	31.5
		MCWB	19.8	20.3	21.3	22.1	23.3	24.4	24.1	24.3	24.4	22.1	21.1	20.9
	2%	DB	30.6	34.1	37.7	40.1	40.8	38.1	32.0	30.8	31.7	33.1	31.4	30.3
		MCWB	19.3	19.9	20.8	22.7	22.9	24.3	24.1	24.0	24.0	22.0	21.0	19.9
	5%	DB	29.5	32.8	36.5	39.2	39.9	36.2	30.8	29.8	30.7	32.1	30.6	29.3
		MCWB	19.1	19.8	20.6	22.4	22.7	24.1	24.1	23.8	23.7	21.6	20.8	19.3
	10%	DB	28.4	31.5	35.3	38.2	38.9	34.4	29.6	28.8	29.7	31.2	29.7	28.2
		MCWB	18.6	19.8	20.3	22.0	22.8	24.1	23.9	23.7	23.7	21.4	20.5	18.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.6	24.3	25.7	28.5	28.7	27.4	25.7	25.7	26.2	25.3	24.4	23.1
		MCDB	29.2	32.1	35.1	37.7	38.4	35.6	29.8	29.8	31.1	30.3	29.2	29.7
	2%	WB	21.3	22.8	24.2	26.5	27.1	26.2	25.1	25.0	25.2	24.6	23.5	21.7
		MCDB	28.4	31.0	34.3	37.2	36.3	33.1	29.3	28.9	29.5	29.3	28.5	27.9
	5%	WB	20.2	21.7	23.0	25.0	25.9	25.5	24.7	24.5	24.7	24.0	22.8	20.7
		MCDB	27.7	30.3	33.4	36.2	35.1	32.0	28.7	28.1	28.8	28.5	27.8	27.1
	10%	WB	19.3	20.6	21.7	23.7	24.9	25.0	24.2	24.0	24.3	23.5	22.1	19.8
		MCDB	27.0	29.3	32.6	35.1	34.1	31.3	28.1	27.5	28.1	27.7	27.1	26.4
Mean Daily Temperature Range	5% DB	MDBR	14.4	14.5	14.0	13.2	12.5	9.2	6.2	6.1	7.4	11.1	13.3	14.5
		MCDBR	15.4	15.8	14.7	13.6	13.0	12.0	8.5	7.9	9.1	12.9	14.3	15.3
		MCWBR	7.5	6.9	6.2	5.7	4.5	3.4	2.7	2.9	3.3	4.8	6.5	7.4
	5% WB	MCDBR	13.8	14.1	14.0	13.1	12.1	9.8	6.9	6.7	7.5	8.7	11.0	13.3
		MCWBR	7.9	8.2	7.7	7.4	5.3	3.6	2.9	3.1	3.4	3.8	5.9	7.4
Clear-Sky Solar Irradiance	taub	0.443	0.453	0.490	0.530	0.542	0.584	0.562	0.536	0.523	0.520	0.493	0.449	
	taud	2.079	2.068	1.987	1.933	1.943	1.901	1.965	2.044	2.063	1.996	2.008	2.073	
	Ebn at Noon	823	841	824	794	776	738	754	779	786	770	768	800	
	Edh at Noon	152	162	183	195	191	198	186	172	167	173	162	149	
All-Sky Solar Radiation	RadAvg	4.77	5.64	6.49	7.02	7.14	5.35	4.00	3.98	4.75	5.18	4.75	4.57	
	RadStd	0.15	0.19	0.27	0.17	0.24	0.49	0.38	0.29	0.34	0.27	0.28	0.15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.45	+1.05	N/A	+0.52	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	+0.45	+1.05	N/A	+0.52	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page